Analytical Cumulative Examination
October 5, 1996

1. (40 points) Consider a weak acid, HA which has an acid dissociation constant, K_a. At a specific wavelength, the molecular form of the acid, HA has a molar absorptivity of ε_{HA}, while the anion of the acid, A^- has a molar absorptivity of ε_{A^-}. The analytical concentration (total A species) is given by C_A. Derive an expression for the absorbance of a solution of HA in terms of C_A, K_a, $[H^+]$, ε_{HA} and ε_{A^-}. From your analysis, indicate how the Beer's law plot depends on the relative magnitude of ε_{HA} to ε_{A^-}, the pH and C_A. The $[H^+]$ can be independently adjusted by using another buffer.

2. (30 points) Picric acid (2,4,6-trinitrophenol) reacts with amines, with a stoichiometry of 1 picric acid per amine functionality. The molar absorptivity of the picric acid/amine complex is virtually identical to that of picric acid. An experiment is carried out to estimate the molecular weight of an unknown amine. 25.6 mg of picric acid is dissolved in 100 ml of ethanol. 1.00 ml of this solution is volumetrically diluted to 100 ml with ethanol. The absorbance of this solution was measured as 0.15 cm^{-1} at 380 nm. A sample of the unknown amine was reacted with an excess of picric acid and the product isolated and purified. 5 mg of the amine/picric acid product was dissolved in ethanol to a volume of 100 ml. This solution had an absorbance of 0.210 cm^{-1} at 380 nm. Determine the molecular weight for the unknown amine assuming it has only one amine functionality per molecule.

3. (30 points) In a flow injection analysis experiment, a sample of 1.5×10^{-4} M Cu^{2+} is injected into a buffered carrier stream containing a reagent that reacts with Cu^{2+} to form a colored product. As the sample is carried downstream toward an absorption detector, the sample mixes with the reagent in a reaction coil (short length of coiled tubing). The product of the reaction is detected spectrophotometrically.

a. Identify as many variables of the system as you can. Indicate those variables which would be difficult to vary experimentally, and those which you would want to control in order to obtain accurate and precise quantitative data.

b. The flowing stream will shape the injection plug into a Gaussian-like distribution. The average height of the sample peak is h, while the width is defined by the standard deviation of the distribution, σ. Comment on the effect of optimizing the response function $R = h/\sigma$. Specifically, how would one independently optimize h and σ? Can σ be made too small?

c. Given your analysis in part b, what would be the advantage of optimizing the function $R = h/(1 + \sigma)$?